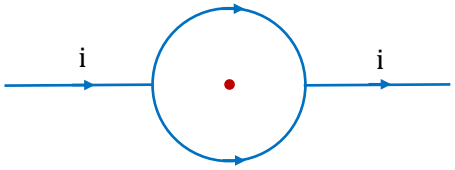
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Biot-Savart Law and Ampere's Law

1. A conducting circular loop of radius a is connected to two long, straight wires. The straight wires carry a current i as shown in figure. Find the magnetic field B at the centre of the loop.



(A) zero (B) $\frac{\mu_0 I}{2a}$ (C) $\frac{\mu_0 I}{a}$ (D) $\frac{\mu_0 I}{a}+\frac{\mu_0 I}{2\pi a}$

PMG084

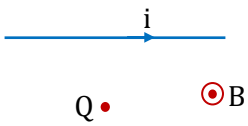
2. A piece of wire carrying a current of 6.00 A is bent in the form of a circular arc of radius 10.0 cm, and it subtends an angle of 120° at the centre. Find the magnetic field B due to this piece of wire at the centre.

(A) zero (B) $1.26 \times 10^{-5}\ T$ (C) $5 \times 10^{-5}\ T$ (D) $7.2 \times 10^{-5}\ T$

PMG085

3. A long, straight wire carrying a current of 1.0 A is placed horizontally in a uniform magnetic field $B = 1.0 \times 10^{-5}\ T$ pointing vertically upward (figure). The magnitude of the resultant magnetic field at the points P and Q, both situated at a distance of 2.0 cm from the wire in the same horizontal plane are respectively

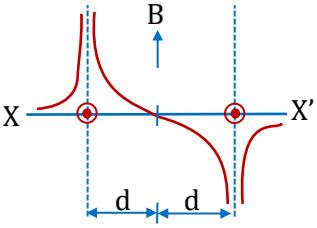
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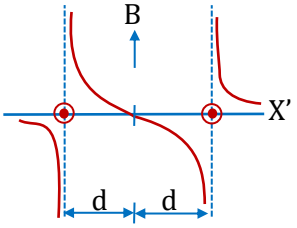


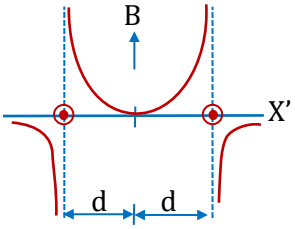
(A) zero, 20 μT (B) 20 μT, zero (C) zero, zero (D) 20 μT, 20 μT

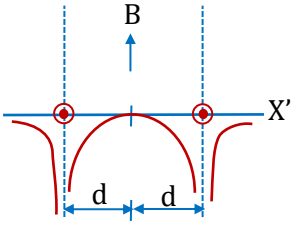
PMG087

4. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the XX' is given by

(A)


(B)


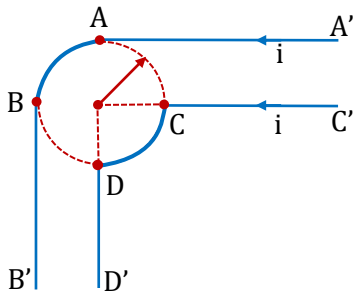
(C)


(D)


PMG088

For More Join: @IITJEE_Advanced

5. All straight wires are very long. Both AB and CD arc area of the same circle, both subtending right angles at the centre O. Then the magnetic field at O is



(A) $\frac{\mu_0 i}{4\pi R}$ (B) $\frac{\mu_0 i}{4\pi R}\sqrt{2}$ (C) $\frac{\mu_0 i}{2\pi R}$ (D) $\frac{\mu_0 i}{2\pi R}(\pi+1)$

PMG089

6. A long straight wire along the z-axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the $z = 0$ plane is :-

(A) $\frac{\mu_0 I}{2\pi}\frac{(y\hat{i}-x\hat{j})}{(x^2+y^2)}$ (B) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{i}+y\hat{j})}{(x^2+y^2)}$ (C) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{j}-y\hat{i})}{(x^2+y^2)}$ (D) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{i}-y\hat{j})}{(x^2+y^2)}$

PMG093

7. A straight wire extends from $(0, 0, -5m)$ to $(0, 0, +5m)$ and carries a current of $2A$ in $+z$ direction. Magnetic field at $(3, 4, 0)$ is

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(A) $\frac{2\sqrt{2}}{25}\times 10^{-7}(-4\hat{i}+3\hat{j})\,T$ (B) $\frac{2\sqrt{2}}{25}\times 10^{-7}(3\hat{i}+4\hat{j})\,T$

(C) $\frac{2\sqrt{2}}{25}\times 10^{-7}(4\hat{i}-3\hat{j})\,T$ (D) None

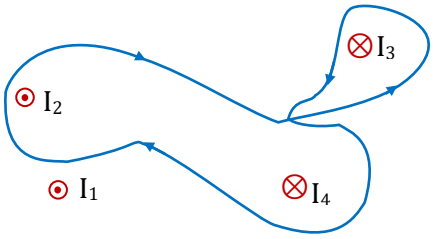
PMG094

8. Considering magnetic field along the axis of a circular loop of radius 1 meter, at what distance from the centre of the loop the magnetic field is $\frac{1}{2\sqrt{2}}$ times of its value at the centre ?

(A) 1 m (B) 3 m (C) 5 m (D) 9 m

PMG097

9. Four wires carrying current $I_1 = 2A$, $I_2 = 4A$, $I_3 = 6A$ and $I_4 = 8A$ respectively cut the page perpendicularly as shown in figure. The value of $\int \vec{B}.d\vec{\ell}$ for the loop shown would be :-

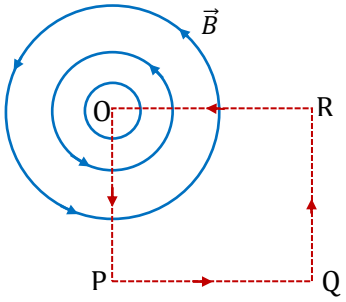


(A) +2 μ_0-wb/m (B) −2 μ_0-wb/m (C) + 10 μ_0-wb/m (D) − 10 μ_0-wb/m

PMG101

For More Join: @IITJEE_Advanced

10. A current carrying wire (current = i) perpendicular to the plane of the paper produces a magnetic field, as shown in the figure. A square of side a is drawn with one of its vertices on the centre of the wire. The integral $\int \vec{B}.d\vec{r}$ along $OPQRO$ has the value



(A) $+\mu_0 i$ (B) $\frac{\mu_0 i}{8}$ (C) $\frac{\mu_0 i}{4}$ (D) $\frac{\mu_0 i}{2}$

PMG103

11. A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross-section. The ratio of the magnetic field at $\frac{a}{2}$ and $2a$ is-

(A) $\frac{1}{4}$ (B) 4 (C) 1 (D) 2

PMG104

12. **Statement-1 :** Ampere law is useful to find magnetic field due to finite length of a straight current carrying wire.

Statement-2 : The magnetic field due to finite length of a straight current carrying wire is symmetric about the wire.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

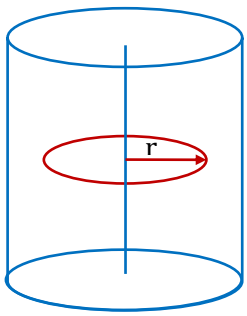
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

PMG105

13. The magnetic field inside a solid conducting long wire at distance r from its axis is given as $B = B_0 r^3$ where B_0 is constant. Which of the following relations correctly represents current enclosed in the loop of radius r shown in the figure ?



(A) $\frac{2\pi B_0 r^3}{5\mu_0}$ (B) $\frac{\pi B_0 r^4}{2\mu_0}$ (C) $\frac{2\pi B_0 r^2}{\mu_0}$ (D) $\frac{2\pi B_0 r^4}{\mu_0}$

PMG106

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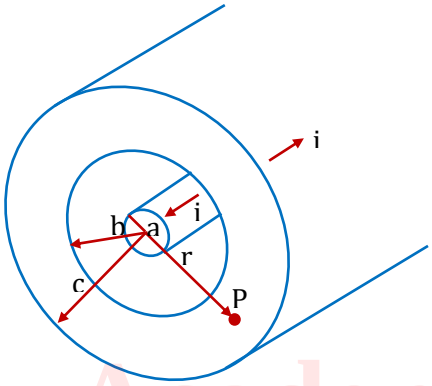
14. A long solenoid has 200 turns per *cm* and carries a current *i*. The magnetic field at its centre is 6.28×10^{-2} weber/m^2. Another long solenoid has 100 turns per *cm* and it carries a current $i/3$. The value of the magnetic field at its centre is-

(A) 1.05×10^{-2} weber/m^2 (B) 1.05×10^{-5} weber/m^2

(C) 1.05×10^{-3} weber/m^2 (D) 1.05×10^{-4} weber/m^2

PMG107

15. A coaxial cable having radius "*a*" of inner wire and inner and outer radii "*b*" and "*c*" respectively of the outer shell carries equal and opposite currents of magnitude *i* on the inner and outer conductors as shown. What is the magnitude of the magnetic induction at point *P* of the cable at a distance *r* ($b < r < c$) from the axis?

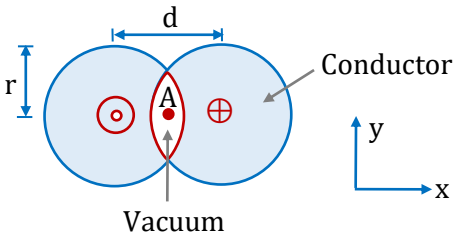


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(A) Zero (B) $\frac{\mu_0 ir}{2\pi a^2}$ (C) $\frac{\mu_0 i}{2\pi r}$ (D) $\frac{\mu_0 i}{2\pi r}\frac{c^2-r^2}{c^2-b^2}$

PMG111

16. Two long conductors are arranged as shown above to form overlapping cylinders, each of radius *r*, whose centers are separated by a distance *d*. Current of density *J* flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point *A*?



(A) $(\mu_0/2\pi)\pi dJ$, in the +*y*-direction (B) $(\mu_0/2\pi)d^2/r$, in the +*y*-direction

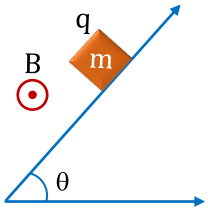
(C) $(\mu_0/2\pi)4d^2J/r$, in the –*y*-direction (D) $(\mu_0/2\pi)Jr^2/d$, in the –*y*-direction

PMG113

For More Join: @IITJEE_Advanced

Magnetic Force on a Charged Particle

17. A block of mass m and charge q is released on a long smooth inclined plane magnetic field B is constant, uniform, horizontal and parallel to surface as shown. Find the time from start when block loses contact with the surface.



(A) $\frac{m\cos\theta}{qB}$ (B) $\frac{m\ \text{cosec}\ \theta}{qB}$ (C) $\frac{m\cot\theta}{qB}$ (D) none

PMG115

18. A charged particle enters a non-uniform uni-directional field such that initial velocity is parallel to magnetic field, then the radius of curvature of its path is (in standard notation):

(A) mV/qB (B) 0 (C) ∞ (D) qB/mV

PMG117

19. An electron moving with a velocity $\vec{V}_1 = 2\hat{\imath}m/s$ at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{\jmath}N$. If the electron is moving with a velocity $\vec{V}_2 = 2\hat{\jmath}m/s$ at the same point, it experiences a force $\vec{F}_2 = +2\hat{\jmath}N$. The force the electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}m/s$ at the same point is

(A) zero (B) $2\hat{k}N$

(C) $-2\hat{k}N$ (D) information is insufficient

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PMG120

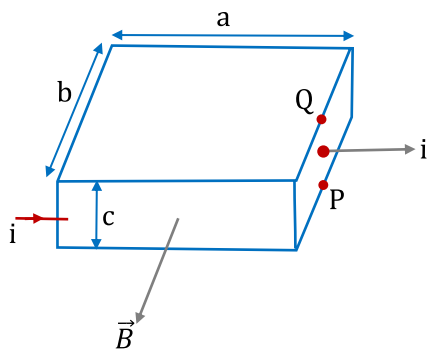
20. A charged particle moves in a magnetic field $\vec{B} = 10\hat{\imath}$ with initial velocity $\vec{u} = 5\hat{\imath} + 4\hat{\jmath}$. The path of the particle will be

(A) straight line (B) circle (C) helical (D) none

PMG121

21. A current flows through a rectangular conductor in the presence of uniform magnetic field B pointing out of the page as shown. Then the potential difference $V_P - V_Q$ is equal to

(Assume charge carriers in the conductor to be positively charged moving with speed v)



(A) Bvc (B) $-Bvc$ (C) Bvb (D) $-Bvb$

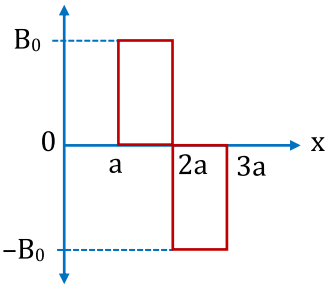
PMG123

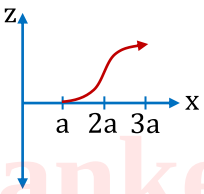
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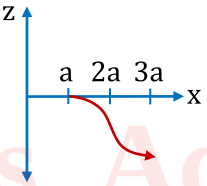
22. In a cyclotron, a charged particle:
(A) undergoes acceleration all the time.
(B) speeds up between the dees because of the magnetic field.
(C) speeds up in a dee.
(D) slows down within a dee and speeds up between dees.

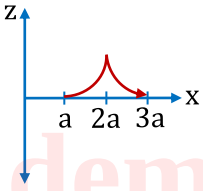
PMG124

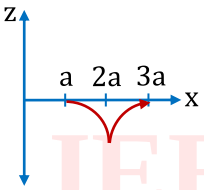
23. A magnetic field $\vec{B} = B_0\,\hat{j}$ exists in the region $a < x < 2a$ and $\vec{B} = -B_0\hat{j}$, in the region $2a < x < 3a$, where B_0 is a positive constant. A positive point charge moving with a velocity = v_0, where v_0 is a positive constant, enters the magnetic field at $x = a$. The trajectory of the charge in this region can be like,



(A) 

(B) 

(C) 

(D) 

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PMG125

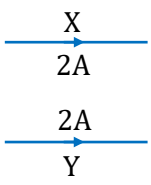
24. A particle of specific charge (charge/mass) α starts moving from the origin under the action of an electric field $\vec{E} = \vec{E}_0\hat{\imath}$ and magnetic field $\vec{B} = B_0\hat{k}$. Its velocity at $(x_0, y_0, 0)$is $(4\hat{\imath} - 3\hat{\jmath})$. The value of x_0 is:

(A) $\frac{13}{2}\frac{\alpha E_0}{B_0}$ (B) $\frac{16\alpha B_0}{E_0}$ (C) $\frac{25}{2\alpha E_0}$ (D) $\frac{5\alpha}{2B_0}$

PMG126

Magnetic Force on a Current Carrying Wire

25. In given figure, *X* and *Y* are two long straight parallel conductors each carrying a current of 2 *A*. The force on each conductor is *F* newtons. When the current in each is changed to 1 *A* and reversed in direction, the force on each is now



(A) $F/4$ and unchanged in direction
(B) $F/2$ and reversed in direction
(C) $F/2$ and unchanged in direction
(D) $F/4$ and reversed in direction

PMG128

For More Join: @IITJEE_Advanced

26. Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A.

(A) 2×10^{-5} N ; attractive force normal to B towards A.

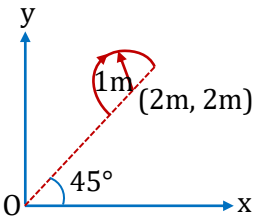
(B) 2×10^{-5} N ; attractive force normal to A towards B.

(C) 5×10^{-5} N ; attractive force normal to A towards B.

(D) 5×10^{-5} N ; attractive force normal to B towards A.

PMG129

27. A uniform magnetic field $\vec{B} = (3\hat{\imath} + 4\hat{\jmath} + \hat{k})$ Tesla exist in a region of space. A semicircular wire of radius 1 m carrying a current of 1A having its centre at (2, 2, 0) m is placed on the X-Y plane as shown. The force on the semi-circular wire will be

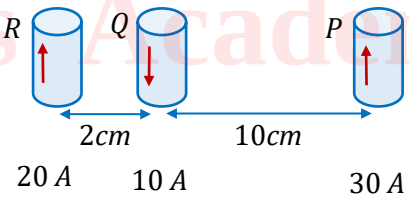


(A) $(\hat{\imath} - \hat{\jmath} + \hat{k})N$ (B) $\sqrt{2}(\hat{\imath} - \hat{\jmath} + \hat{k})N$ (C) $\frac{1}{\sqrt{2}}(\hat{\imath} + \hat{\jmath} - \hat{k})N$ (D) $\sqrt{2}(\hat{\imath} + \hat{\jmath} - \hat{k})N$

PMG130

28. Three long, straight and parallel wires carrying currents are arranged as shown in figure. The force experienced by 10 cm length of wire Q is

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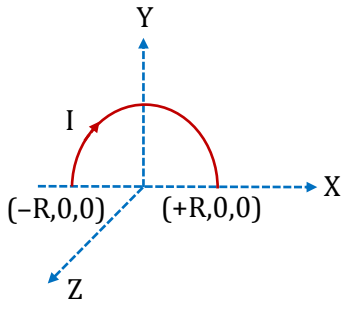


(A) $1.4 \times 10^{-4} N$ towards the right (B) $1.4 \times 10^{-4} N$ towards the left

(C) $2.6 \times 10^{-4} N$ to the right (D) $2.6 \times 10^{-4} N$ to the left

PMG131

29. A semi circular current carrying wire having radius R is placed in x-y plane with its centre at origin 'O'. There is non-uniform magnetic field $\vec{B} = \frac{B_o x}{2R}\hat{k}$ (here B_o is +ve constant) is existing in the region. The magnetic force acting on semi circular wire will be along :-

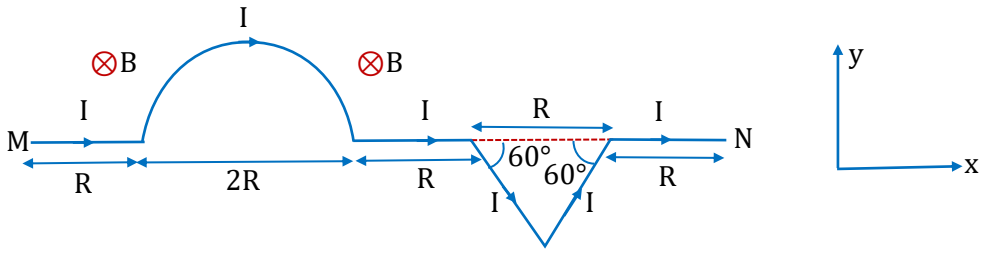


(A) – x-axis (B) + y-axis (C) – y-axis (D) + x-axis

PMG132

For More Join: @IITJEE_Advanced

30. A conducting wire *MN* carrying a current *I* is bent into the shape as shown and placed in *xy* plane. A uniform magnetic field $\vec{B} = -B\hat{k}$ is existing in the region. The net magnetic force experienced by the conducting wire *MN* is :-

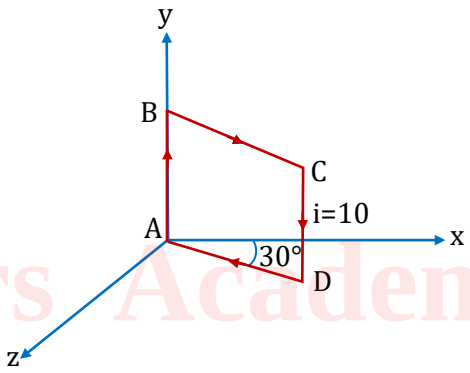


(A) 2 *BIR* (B) 3 *BIR* (C) 4 *BIR* (D) 6 *BIR*

PMG135

Torque on a Current Carrying Loop

31. Figure shows a square current carrying loop *ABCD* of side 10 *cm* and current *i* = 10*A*. The magnetic moment $\vec{M}$ of the loop is



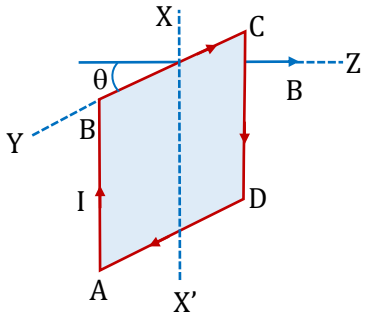
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(A) $(0.05)\left(\hat{\imath} - \sqrt{3}\hat{k}\right)A - m^2$ (B) $(0.05)\left(\hat{\jmath} + \hat{k}\right)A - m^2$

(C) $(0.05)\left(\hat{\jmath} + \hat{k}\right)A - m^2$ (D) $\left(\hat{\imath} + \hat{k}\right)A - m^2$

PMG138

32. The square loop *ABCD*, carrying a current *I*, is placed in a uniform magnetic field *B*, as shown. The loop can rotate about the axis *XX'*. The plane of the loop makes an angle θ (θ < 90°) with the direction of *B*. Through what angle will the loop rotate by itself before the torque on it becomes zero–

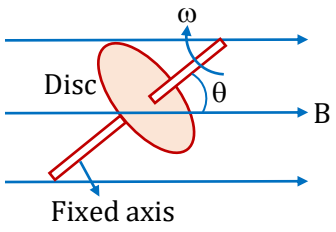


(A) θ (B) 90° – θ (C) 90° + θ (D) 180° – θ

PMG139

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33. A disc of radius r and carrying positive charge q is rotating with an angular speed ω in a uniform magnetic field B about a fixed axis as shown in figure, such that angle made by axis of disc with magnetic field is θ. Torque applied by axis on the disc is

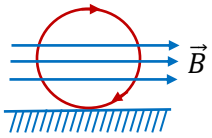


(A) $\frac{q\omega r^2 B\sin\theta}{2}$, clockwise (B) $\frac{q\omega r^2 B\sin\theta}{4}$, anticlockwise

(C) $\frac{q\omega r^2 B\sin\theta}{2}$, anticlockwise (D) $\frac{q\omega r^2 B\sin\theta}{4}$, clockwise

PMG140

34. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current $i = 4A$. A horizontal magnetic field $B = 10T$ is switched on at time $t = 0$ as shown in figure. The initial angular acceleration of the ring will be :-

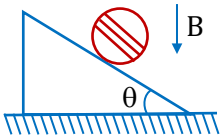


(A) 40 π rad/s^2 (B) 20 π rad/s^2 (C) 5 π rad/s^2 (D) 15 π rad/s^2

PMG141

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35. In the figure shown a coil of single turn is wound on a sphere of radius R and mass m. The plane of the coil is parallel to the plane and lies in the equatorial plane of the sphere. Current in the coil is i. The value of B if the sphere is in equilibrium is :-



(A) $\frac{mg\cos\theta}{\pi iR}$ (B) $\frac{mg}{\pi iR}$ (C) $\frac{mg\tan\theta}{\pi iR}$ (D) $\frac{mg\sin\theta}{\pi iR}$

PMG142

36. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 divisions per milli ampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 V, the resistance in Ohm needed to be connected in series with the coil will be-

(A) 10^3 (B) 10^5 (C) 99995 (D) 9995

PMG145

Magnetic Materials and Earth's Magnetism

37. The magnetic lines of force inside a bar magnet:

(A) are from north-pole to south-pole of the magnet

(B) do not exist

(C) depend upon the area of cross-section of the bar magnet

(D) are from south-pole to north-pole of the magnet

PMG150

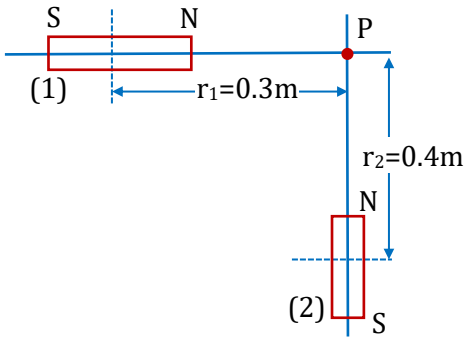
For More Join: @IITJEE_Advanced

38. A steel wire of length L has a magnetic moment M. It is then bent into a semi-circular arc; the new magnetic moment will be :

(A) M (B) $2M/\pi$ (C) M/L (D) $M \times L$

PMG151

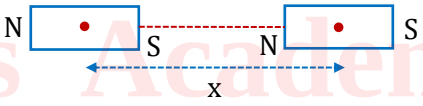
39. Two short magnets of magnetic moment 2 Am^2 and 5 Am^2 respectively are placed along two lines drawn at right angle to each other on the sheet of paper as shown in the figure. What is the magnetic field at the point of intersection of their axis:



(A) $2.15 \times 10^{-5}\ T$ (B) $215 \times 10^{-5}\ T$ (C) $2.15 \times 10^{-4}\ T$ (D) $21.5 \times 10^{-5}\ T$

PMG152

40. The mid points of two small magnetic dipoles of length d in end-on positions, are separated by a distance x, ($x >> d$). The force between them is proportional to x^{-n} where n is :-



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(A) 3 (B) 4 (C) 2 (D) 1

PMG154

41. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is 2s. The magnet is cut along its length into three equal parts and three parts are then placed on each other with their like poles together. The time period of this combination will be-

(A) 2s (B) 2/3 s (C) $2\sqrt{3}\ s$ (D) $2/\sqrt{3}\ s$

PMG161

42. A specimen of iron of permeability 8×10^{-3} weber/amp × metre is placed in a magnetic field of intensity 160 amp/metre. Then magnetic field in this iron is :

(A) $20 \times 10^3\ Wb/m^2$ (B) $1.28\ Wb/m^2$ (C) $5 \times 10^{-5}\ Wb/m^2$ (D) $0.8\ Wb/m^2$

PMG162

43. Curie temperature is the temperature above which :

(A) a paramagnetic material becomes diamagnetic

(B) a ferromagnetic material becomes diamagnetic

(C) a paramagnetic material becomes ferromagnetic

(D) a ferromagnetic material becomes paramagnetic

PMG163

For More Join: @IITJEE_Advanced

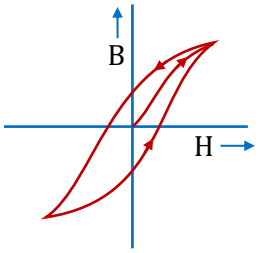
44. If a solution of ferromagnetic material is poured into a *U*-tube and one arm of this is placed between the poles of a strong magnet with the meniscus in line with the field, then the level of the solution will:

(A) Rise (B) Fall

(C) Rise till the liquid comes out (D) Remain unchanged

PMG164

45. The *B*-*H* curve for a certain specimen is schematically shown by the given diagram. Which one of the following is the correct magnetic nature of the specimen?



(A) Diamagnetic and not ferromagnetic or paramagnetic

(B) Ferromagnetic and not diamagnetic or paramagnetic

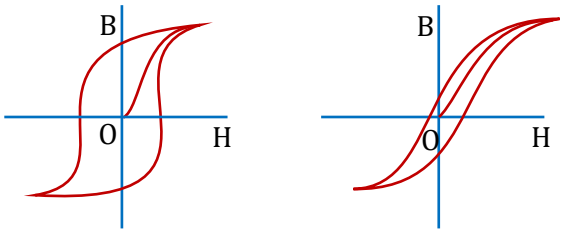
(C) Paramagnetic and not diamagnetic or ferromagnetic

(D) Applicable to all the three types of magnetism mentioned above

PMG166

Rankers Academy JEE

46. Two types of steel are characterized by the hysteresis loops shown in figure (a) and (b). The loops are obtained in the processes of magnetization and demagnetization of the steels. Which of the two types is best suited for using as the core of a transformer and which, for using as a permanent magnet ?



(a) **(b)**

(A) Both are best suited for transformer

(B) Both are best suited for permanent magnet

(C) *A* is best suited for permanent magnet and *B is* best suited for core of transformer

(D) *A* is best suited for core of transformer and *B* is best suited for permanent magnet

PMG167

47. A material satisfied the relation $\mu_0(H + M) = 0$, where H and M are magnetic intensity and magnetisation respectively, then material is :-

(A) nonmagnetic (B) paramagnetic (C) ferromagnetic (D) diamagnetic

PMG168

For More Join: @IITJEE_Advanced

48. **Statement-1 :** The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil.

and

Statement-2 : Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.

(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True; statement-2 is NOT a correct explanation for statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

PMG170

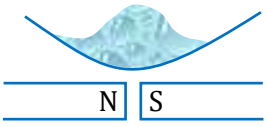
49. The materials suitable for making electromagnets should have-

(A) high retentivity and high coercivity (B) low retentivity and low coercivity

(C) high retentivity and low coercivity (D) low retentivity and high coercivity

PMG171

50. The liquid in the watch glass in the following figure is:



(A) Ferromagnetic (B) Paramagnetic (C) Diamagnetic (D) Nonmagnetic

PMG172

Rankers Academy JEE

51. The magnetic susceptibility of a paramagnetic material at $-73°C$ is 0.0075 then its value at $-173°C$ will be :-

(A) 0.0045 (B) 0.0030 (C) 0.015 (D) 0.0075

PMG173

52. Which statement is true:

(A) atomic magnetic dipole moment of diamagnet is zero

(B) atomic magnetic dipole moment of para-magnet is zero

(C) atomic magnetic dipole moment of ferro-magnet is zero

(D) ferromagnet is demagnetised rapidly after moving in magnetising field.

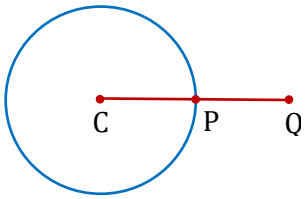
PMG175

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

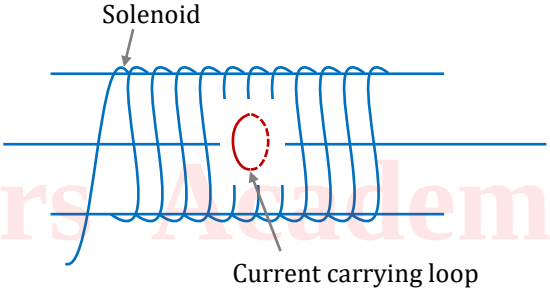
1. An electron is at P at $t = 0$. It is circulating in anti clockwise direction with a constant angular speed ω along the shown circular path. Magnetic field at $Q(CQ = 2R)$, where R is radius of the circle) will be recorded as zero at times.



(A) $\frac{\pi}{3\omega}$ (B) $\frac{\pi}{\omega}$ (C) $\frac{5\pi}{3\omega}$ (D) $\frac{7\pi}{3\omega}$

PMG099

2. A single circular loop of wire with radius 0.02 m carries a current of 8.0 A. It is placed at the centre of a long solenoid that has length 0.65m, radius 0.080 m and 1300 turns. (Axis of circular loop coincide with axis of solenoid). Let 'i' be the current in solenoid.



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(A) If $i = 100\ mA$, the magnetic field at the center of coil can be zero.

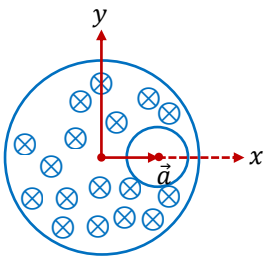
(B) If $i = 44\ mA$, the magnetic field at the center of coil can be zero.

(C) If $i = 100\ mA$, the magnetic field at the center of coil can be $8\pi \times 10^{-5}\ T$.

(D) If $i = 100\ mA$, the magnetic field at the center of coil can be $16\pi \times 10^{-5}\ T$.

PMG108

3. Figure shows cross-section view of an infinite cylindrical wire with a cylindrical cavity, current density is uniform $\vec{J} = -j_0\hat{k}$ as shown in figure.



(A) Field inside cavity is uniform.

(B) Field inside cavity is along $\vec{a}$.

(C) Field inside cavity is perpendicular to $\vec{a}$.

(D) If an electron is projected with velocity $v_0\,\vec{j}$ it will move undeviated before colliding with cavity wall.

PMG112

For More Join: @IITJEE_Advanced

4. A long cylindrical conductor whose axis is coincide with the z axis has an internal magnetic field given by

$$B=\frac{\mu_0 J_0 r}{6a}(3a-2r)$$

where a is the conductor's radius.

(A) The total current flowing in the conductor is $\frac{J_0\pi a^2}{3}$.

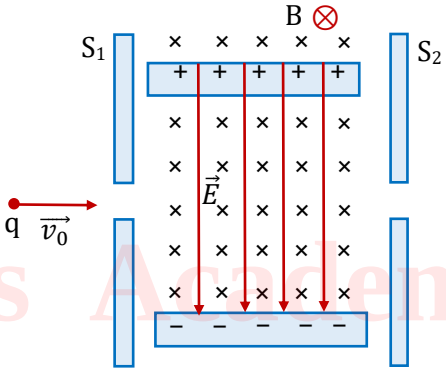
(B) The magnetic field outside the conductor is $\frac{\mu_0 J_0 \pi a^2}{6r}$.

(C) The current density is maximum at the axis of the conductor.

(D) The current density is minimum at the axis of the conductor.

PMG055

5. A positive charge enters a region of perpendicular electric and magnetic fields as shown in the figure, travelling with speed V_0 directed horizontally to the right. Initially the charge moves through the region shown with no deflection and emerges through the hole in screen S_2. If we now want the charge to hit below the hole in the screen S_2 we should not :-



Rankers Academy JEE

(A) Decrease its speed below V_0 or increase the magnitude of E.

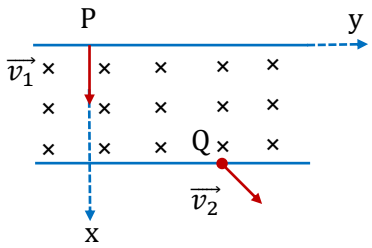
(B) Decrease its speed below V_0 or decrease the magnitude of E.

(C) Increase its speed above V_0 or increase the magnitude of E.

(D) Increase its speed above V_0 or decrease the magnitude of E.

PMG127

6. A charged particle moving in x-y plane enters a region of uniform magnetic field of intensity $\vec{B}=-B_0\hat{k}$ at $P(0, 0)$ with velocity $\vec{v}_1=v\hat{\imath}$ and leaves the magnetic field at $Q(a, b)$ with velocity $\vec{v}_2=\frac{v}{2}\hat{\imath}+\frac{\sqrt{3}}{2}v\hat{\jmath}$. Choose the correct statement(s)



(A) The particle can't be positron

(B) Deviation suffered by the particle as it moves from P to Q equals 60°

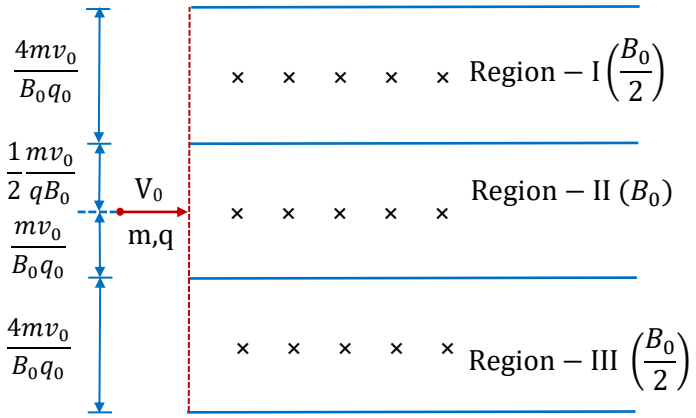
(C) $\sqrt{3}a=b$

(D) $\vec{v}_1=\vec{v}_2$

PMG056

For More Join: @IITJEE_Advanced

7. In given figure at $t = 0$, a charged particle of mass m and charge q enters in the region-II.



(A) If $q > 0$, particle will enter into region-I after leaving region-II.

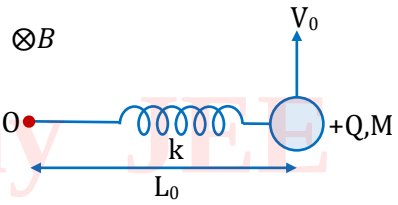
(B) If $q < 0$ the particle will leave region-II at $t = \frac{\pi m}{2|q|B_0}$

(C) If $q > 0$ particle will re-enter in region-II at $t = \frac{\pi m}{3B_0q}$

(D) Particle will not leave region-II.

PMG057

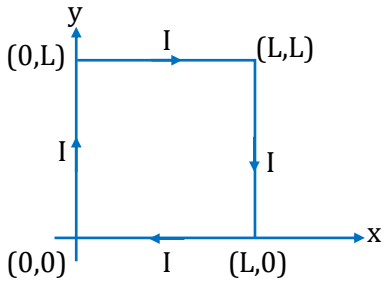
8. In zero gravity region, a point charge $+Q$ of mass M is connected to a spring of natural length L_0 and spring constant k. A uniform magnetic field B is directed inside the plane of paper. The point charge is given an initial velocity v_0 perpendicular to the length of spring as shown. It is given that $B = \frac{Mv_0}{QL_0}$. Then which of the following quantities remain(s) conserved during the motion of particle?



(A) Kinetic energy
(B) Potential energy
(C) Total energy of system
(D) Angular momentum of particle about O

PMG178

9. Figure shows a square loop carrying current I is present in the magnetic field which is given by $\vec{B} = \frac{B_0z}{L}\hat{j} + \frac{B_0y}{L}\hat{k}$ where B_0 is positive constant. Which of the following statement(s) is/are **CORRECT**?



(A) force on side (0, 0) to (0, L) is $\left(\frac{B_0IL}{2}\right)\hat{i}$.
(B) force on side (0, L) to (L, L) is $-B_0I\,L\,\hat{j}$.

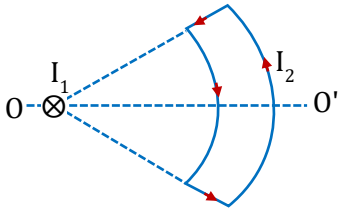
(C) Net magnetic force on loop is zero.
(D) force on side (L, 0) to (0, 0) is zero.

PMG179

For More Join: @IITJEE_Advanced

10. Which of the following statement is correct in the given figure?

Infinitely long wire kept perpendicular to the paper currying current inwards



(A) net force on the loop is zero

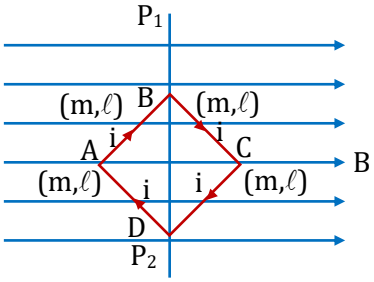
(B) net torque on the loop is zero

(C) loop will rotate clockwise about axis *OO'* when seen from *O*.

(D) loop will rotate anticlockwise about *OO'* when seen from *O*.

PMG181

11. Rod *AB*, *BC*, *CD* & *DA* form a square loop having current i, mass and length of each rod is m and ℓ respectively, is situated in a uniform magnetic field B as shown in the figure and it can rotate about axis P_1P_2. At $t = 0$, if square loop is released from rest from the given position as shown in the diagram, then



(A) Angular acceleration of the square loop at $t = 0$, is $\frac{3}{2}\frac{Bi}{m}$

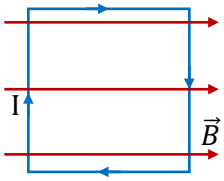
(B) Angular velocity when square loop rotated by 30°, is $\sqrt{\frac{3iB}{2m}}$

(C) Torque on the loop when it rotated by 60°, is $\frac{iB\ell^2}{2}$

(D) Angular acceleration of the square loop, when it is rotated from starting to 90°, decreases continuously

PMG182

12. Figure shows a square frame with a current in a uniform magnetic field of magnitude B in direction shown. The sides are $a \times a$. The net magnetic field at the center of the frame is $\sqrt{3}B$ in magnitude.



(A) The torque on the frame is $\frac{\pi a^3 B^2}{2\mu_0}$.

(B) The potential energy of interaction of frame with the magnetic field is zero.

(C) The total magnetic field at center of the loop will be $\left(\sqrt{3}+1\right)B$ if the magnetic dipole moment of the frame is aligned parallel to the magnetic field.

(D) The total magnetic field at center of the loop will be $\left(\sqrt{2}-1\right)B$ if the magnetic dipole moment of the frame is aligned anti parallel to the magnetic field.

PMG183

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For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 13 AND 14

Electric field and magnetic field in region of space are given by and $\vec{E} = E_0\hat{j}$ and $\vec{B} = B_0\hat{j}$. A particle of specific charge α (charge per unit mass) is released from origin with velocity $\vec{V} = V_0\hat{\imath}$.

13. y coordinate of particle when it crosses y-axis for n^{th} time is :-

(A) $\frac{2n^2\pi^2E_0}{B_0^2\alpha}$ (B) $\frac{n^2\pi E_0}{B_0^2\alpha}$ (C) $\frac{2\pi^2E_0}{n^2B_0\alpha}$ (D) 0

PMG059

14. Angle between particle's velocity and y-axis at that moment is :-

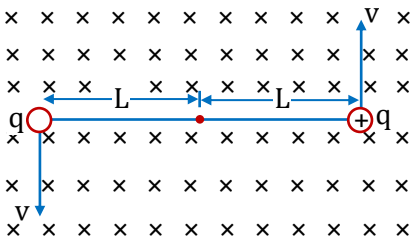
(A) $\tan^{-1}\left(\frac{V_0B_0}{2\pi E_0}\right)$ (B) $\tan^{-1}\left(\frac{V_0B_0}{2\pi nE_0}\right)$ (C) $\tan^{-1}\left(\frac{V_0B_0}{E_0}\right)$ (D) $\tan^{-1}\left(\frac{B_0}{E_0}\right)$

PMG060

PARAGRAPH FOR QUESTION NO. 15 AND 16

Two charge particles each of mass 'm', carrying charge $+q$ and connected with each other by a massless inextensible string of length $2L$ are describing circular path in the plane of paper, each with speed $v = \frac{qB_0L}{m}$ (where B_0 is constant) about their centre of mass in the region in which an uniform magnetic field $\vec{B}$ exists into the plane of paper as shown in figure. Neglect any effect of electrical and gravitational forces.

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15. The magnitude of the magnetic field such that no tension is developed in the string will be

(A) $\frac{B_0}{2}$ (B) B_0 (C) $2B_0$ (D) 0

PMG061

16. If the actual magnitude of magnetic field is half to that of calculated in part (i) then tension in the string will be

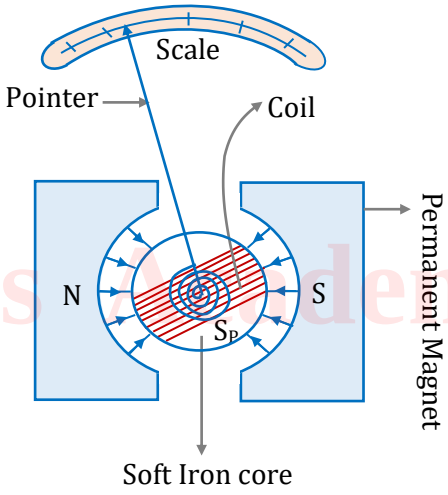
(A) $\frac{3}{4}\frac{q^2B_0^2L}{m}$ (B) zero (C) $\frac{q^2B_0^2L}{2m}$ (D) $\frac{2q^2B_0^2L}{m}$

PMG062

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTIONS NO. 17 TO 19

Analog voltmeters and ammeters work by measuring the torque exerted by a magnetic field on a current carrying coil. The reading is displayed by means of the deflection of a pointer over a scale. The adjacent figure shows the essentials of a galvanometer, on which both analog ammeters and analog voltmeters are based. Assume that the coil is 2.1 *cm* high, 1.2 *cm* wide, has 250 turns and is mounted so that it can rotate about an axis (into the page) in a uniform radial magnetic field with $B = 0.23\ T$. For any orientation of the coil, the net magnetic field through the coil is perpendicular to the normal vector of the coil (and thus parallel to the plane of coil). A spring S_P provides a counter torque that balances the magnetic torque, so that a given steady current I in the coil results in a steady angular deflection ϕ. The greater the current is, greater the deflection is, and thus greater the torque required of the spring is. A current of 100 μA produces an angular deflection of 28°.



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17. What must be the torsional constant K of the spring?

(A) $2.6 \times 10^{-8}\ Nm$/degree (B) $5.2 \times 10^{-8}\ Nm$/degree

(C) $2.6 \times 10^{-4}\ Nm$/degree (D) $5.2 \times 10^{-4}\ Nm$/degree

PMG146

18. If we reduce the value of this K to half of its value, then the deflection would be

(A) 28° (B) 56° (C) 14° (D) none of these

PMG147

19. If the value of magnetic field is put equal to 0.69 T and $K = 15.6 \times 10^{-8}\ Nm$/degree. Then, the deflection would be

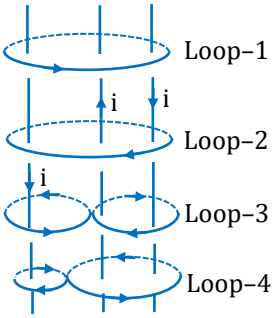
(A) < 28° (B) = 28° (C) 14° (D) none of these

PMG148

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

20. Three wires are carrying same constant current *i* in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in figure.



	Column-I		**Column-II**
(A)	Along closed loop–1	(P)	$\oint \vec{B}.d\vec{\ell} = \mu_0 i$
(B)	Along closed loop–2	(Q)	$\oint \vec{B}.d\vec{\ell} = -\mu_0 i$
(C)	Along closed loop–3	(R)	$\oint \vec{B}.d\vec{\ell} = 0$
(D)	Along closed loop–4	(S)	Net work done by the magnetic force to move a unit charge along the loop is zero

PMG102

Rankers Academy JEE

21. A charge particle with some initial velocity is projected in a region where uniform electric and/or magnetic fields are present. In Column-I information about the existence of electric and/or magnetic field and direction of initial velocity of charge particle are given, while in column-II the possible paths of charged particle is mentioned. Match the entries of Column I with the entries of Column-II.

	Column-I		**Column-II**
(A)	$\vec{E}=0, \vec{B}\neq 0$ and initial velocity is At an unknown angle with $\vec{B}$	(P)	Straight line
(B)	$\vec{E}\neq 0, \vec{B}=0$ and initial velocity is At an unknown angle with $\vec{E}$	(Q)	Parabola
(C)	$\vec{E}\neq 0, \vec{B}\neq 0, \vec{E}\|\|\vec{B}$ and initial velocity Is perpendicular to $\vec{E}$	(R)	Circular
(D)	$\vec{E}\neq 0, \vec{B}\neq 0, \vec{E}$ is perpendicular to $\vec{B}$ And initial velocity is perpendicular Both $\vec{E}$ and $\vec{B}$	(S) (T)	Helical path nonuniform pitch Helical path with uniform pitch

PMG204

For More Join: @IITJEE_Advanced

22. **Column-I (Magnetic moment of)** **Column-II**

(A)	A uniformly charged ring rotating uniformly about its axis	(P)	$\frac{q\omega r^2}{5}$
(B)	A charged particle rotating uniformly about a point	(Q)	$\frac{q\omega r^2}{4}$
(C)	A uniformly charged disk rotating uniformly about its axis	(R)	$\frac{q\omega r^2}{3}$
(D)	A uniformly charged spherical shell rotating uniformly about one of its diameter	(S)	$\frac{q\omega r^2}{2}$
(E)	A uniformly charged sphere rotating	(T)	$q\omega r^2$

PMG149

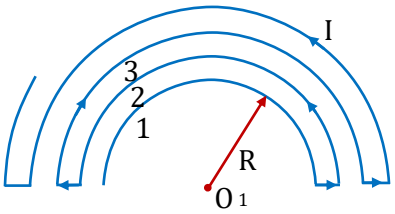
Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

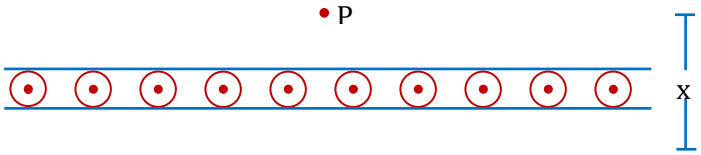
Biot-Savart Law and Ampere's Law

1. Infinite semi-circular wires are placed out from a single wire in a continuous fashion as shown in the figure. The radii of the semicircles increases as $R_n = \alpha^{n-1}R_1$ where R_n is the radius of the nth semicircle and α is constant greater than 1. The semicircles are concentric. The net magnetic field at the centre is given by $\frac{\mu_0}{4}\times\frac{I}{R_1}\left(\frac{\alpha}{\beta+\alpha}\right)$ where β is the numerical constant. Find the value of β.



PMG095

2. Figure shows a cross section of a large metal sheet carrying an electric current along its surface. The current in a strip of width *dl* is *Kdl* where *K* is a constant. Find the magnetic field at a point *P* at a distance *x* from the metal sheet.



PMG109

Rankers Academy JEE

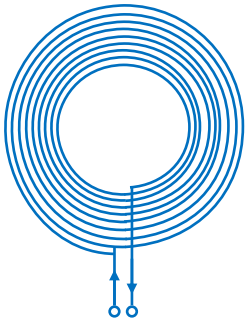
3. An equilateral triangular loop of wire of side ℓ carries a current *i*. The magnetic field produced at the circumcenter of the loop is $\frac{\mu_0}{4\pi}\frac{\alpha i}{\ell}$, then the value of α is

PMG063

4. The current density inside a long solid cylindrical wire of radius *a* is in the direction of the central axis and varies linearly with radial distance *r* from the axis according to $J = J_0\frac{r}{a}$. If the magnetic field inside wire is given by $\frac{\mu_0 J_0 r^x}{ya}$ then find value of $x + y$.

PMG064

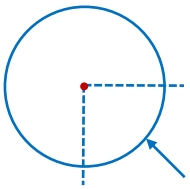
5. A thin insulated wire forms a plane spiral of $N = 100$ tight turns carrying a current $I = 8\ mA$. The radii of inside and outside turns are equal to $a = 50\ mm$ and $b = 100\ mm$. Find the magnetic induction at the centre of the spiral.



PMG184

For More Join: @IITJEE_Advanced

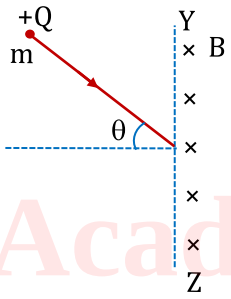
6. A long thin walled pipe of radius R carries a current I along its length. The current density is uniform over the circumference of the pipe. The magnetic field at the center of the pipe due to quarter portion of the pipe shown is



PMG185

Magnetic Force on a Charged Particle

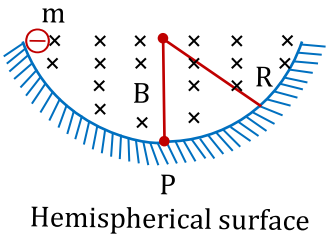
7. A particle with charge $+Q$ and mass m enters a magnetic field of magnitude B, existing only to the right of the boundary YZ. The direction of the motion of the particle is perpendicular to the direction of B. Let $T = 2\pi \frac{m}{QB}$. The time spent by the particle in the field is T_1. In another case if $-Q$ is taken then time spend spent is T_2. Find the ratio of $\frac{T_1}{T_2}$.



Rankers Academy JEE

PMG118

8. A charge particle of mass 'm' and charge $-q$ is released from rest from the given position. In the presence and absence of uniform horizontal magnetic field normal reaction acting on the charge at point P are N_1 and N_2 respectively. Neglect friction. The value of $N_1 - N_2$ is given by $\frac{5Bq\sqrt{2gR}}{k}$. Find k.



Hemispherical surface

PMG122

9. There exists a uniform magnetic and electric field of magnitude 1 T and 1 V/m respectively along positive y-axis. A charged particle of mass 1 kg and charge 1 C moving with velocity 1 m/sec along positive x-axis is at origin at $t = 0$. If the coordinates of particle at time π seconds is given as (X, Y, Z) in meter, then find the value of $(XY + YZ + ZX) \times \frac{5}{\pi^2}$.

PMG067

For More Join: @IITJEE_Advanced

Magnetic Force on a Current Carrying Wire

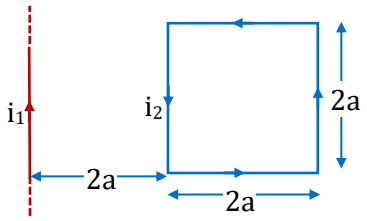
10. A straight horizontal conducting rod of length 0.45 m and mass 60g is suspended by two vertical wires at its ends. A current of 5.0 A is set up in the rod through the wires.

(a) What magnetic field should be set up normal to the conductor in order that the tension in the wires is zero ?

(b) What will be the total tension in the wires if the direction of current is reversed keeping the magnetic field same as before ? [Ignore the mass of the wires] g = 9.8 ms^{-2}.

PMG133

11. An infinitely long straight wire carries a current $i_1 = 1$ amp. A rectangular current loop placed in the same plane of shown dimensions carries a current $i_2 = 1$ amp anticlockwise. Magnitude of net force on the loop is found to be $\beta\frac{\mu_0}{4\pi}$. Find value of β.

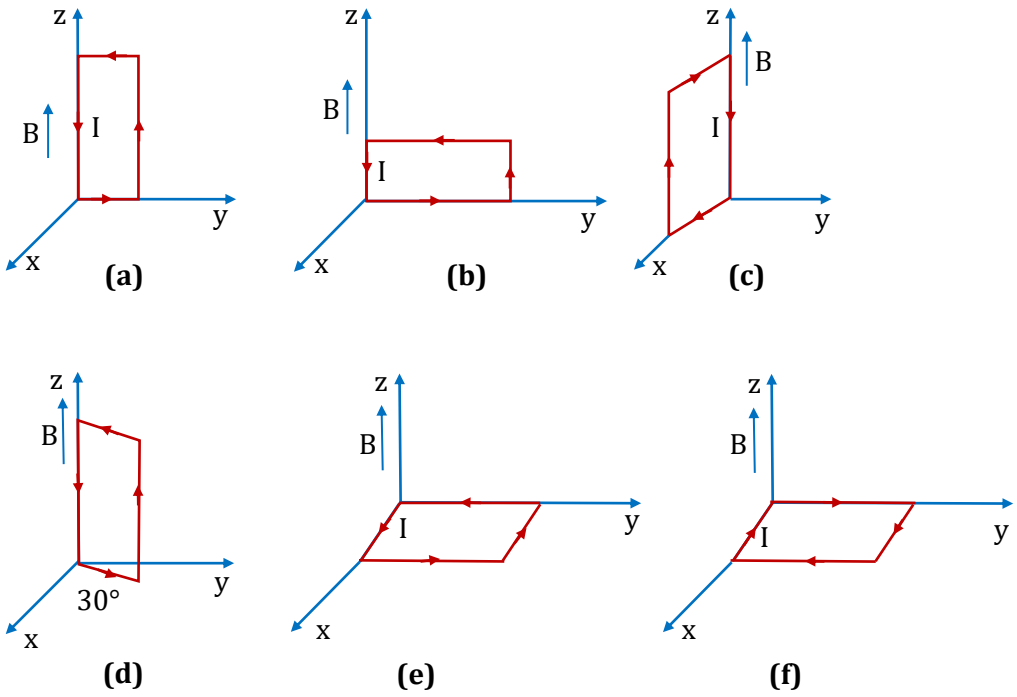


PMG134

Torque on a Current Carrying Loop

12. A uniform magnetic field of 3000 G is established along the positive z-direction. A rectangular loop of sides 10 cm and 5 cm carries a current of 12A. What is the torque on the loop in the different cases shown in figure. What is the force on each case? Which case corresponds to stable equilibrium?

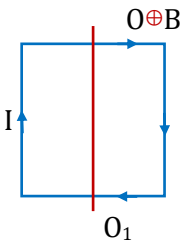
Rankers Academy JEE



PMG143

For More Join: @IITJEE_Advanced

13. A square current carrying loop made of thin wire and having a mass $m = 10g$ can rotate without friction with respect to the vertical axis OO_1, passing through the centre of the loop at right angles to two opposite sides of the loop. The loop is placed in a homogeneous magnetic field with an induction $B = 10^{-1}\,T$ directed at right angles to the plane of the drawing. A current $I = 2A$ is flowing in the loop. Find the period of small oscillations that the loop performs about its position of stable equilibrium.



PMG144

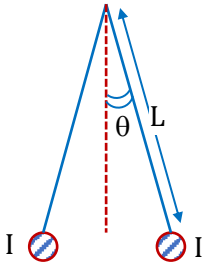
Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Two long current carrying thin wires, both with current I, are held by the insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle 'θ' with the vertical. If wires have mass λ per unit length then the value of I is :- (g = gravitational acceleration)

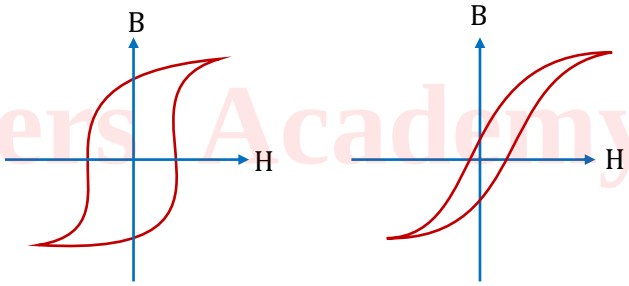
[JEE (Main) 2015]



(1) $2\sqrt{\frac{\pi gL}{\mu_0}\tan\theta}$ (2) $\sqrt{\frac{\pi\lambda gL}{\mu_0}\tan\theta}$ (3) $\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$ (4) $2\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$

PMG032

2. Hysteresis loops for two magnetic materials A and B are given below :



(A) (B)

Rankers Academy JEE

These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use : **[JEE (Main) 2016]**

(1) B for electromagnets and transformers.

(2) A for electric generators and transformers.

(3) A for electromagnets and B for electric transformers.

(4) A for transformers and B for electric generators.

PMG034

3. Two identical wires A and B, each of length 'l', carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side 'a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is : **[JEE (Main) 2016]**

(1) $\frac{\pi^2}{8\sqrt{2}}$ (2) $\frac{\pi^2}{8}$ (3) $\frac{\pi^2}{16\sqrt{2}}$ (4) $\frac{\pi^2}{16}$

PMG035

For More Join: @IITJEE_Advanced

4. A magnetic needle of magnetic moment $6.7 \times 10^{-2}\ Am^2$ and moment of inertia $7.5 \times 10^{-6}\ kg\ m^2$ is performing simple harmonic oscillations in a magnetic field of 0.01 T. Time taken for 10 complete oscillations is : **[JEE (Main) 2017]**

(1) 6.98 s (2) 8.76 s (3) 6.65 s (4) 8.89 s

PMG036

5. The dipole moment of a circular loop carrying a current I, is m and the magnetic field at the centre of the loop is B_1. When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of the loop is B_2. The ratio $\frac{B_1}{B_2}$ is : **[JEE (Main) 2018]**

(1) $\sqrt{3}$ (2) $\sqrt{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 2

PMG037

6. A paramagnetic material has 10^{28} atoms/m^3. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4}. Its susceptibility at 300 K is : **[JEE (Main) 2019]**

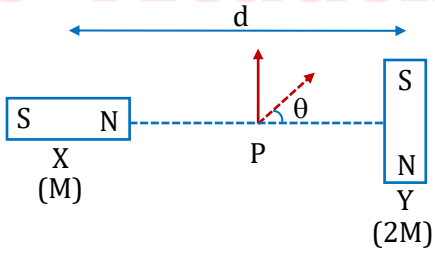
(1) 3.672×10^{-4} (2) 3.726×10^{-4} (3) 3.267×10^{-4} (4) 2.672×10^{-4}

PMG038

7. Two magnetic dipoles X and Y are placed at a separation d, with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing, through their midpoint P, at angle $\theta = 45°$ with the horizontal line, as shown in figure. What would be the magnitude of force on the particle at that instant ?

(d is much larger than the dimensions of the dipole) **[JEE (Main) 2019]**

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(1) $\sqrt{2}\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3}\times qv$ (2) $\left(\frac{\mu_0}{4\pi}\right)\frac{2M}{(d/2)^3}\times qv$

(3) $\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3}\times qv$ (4) 0

PMG039

8. An insulating thin rod of length ℓ has a x linear charge density $p(x) = \rho_0 \frac{x}{\ell}$ on it. The rod is rotated about an axis passing through the origin ($x = 0$) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is : **[JEE (Main) 2019]**

(1) $\frac{\pi}{4}n\rho\ell^3$ (2) $n\rho\ell^3$ (3) $\pi n\rho\ell^3$ (4) $\frac{\pi}{3}n\rho\ell^3$

PMG041

For More Join: @IITJEE_Advanced

9. A particle of mass m and charge q is in an electric and magnetic field given by

$\vec{E} = 2\hat{\imath} + 3\hat{\jmath}\,;\, \vec{B} = 4\hat{\jmath} + 6\hat{k}$

The charged particle is shifted from the origin to the point $P(x = 1\,;\, y = 1)$ along a straight path. The magnitude of the total work done is :- **[JEE (Main) 2019]**

(1) $(0.35)q$ (2) $(0.15)q$ (3) $(2.5)q$ (4) $5q$

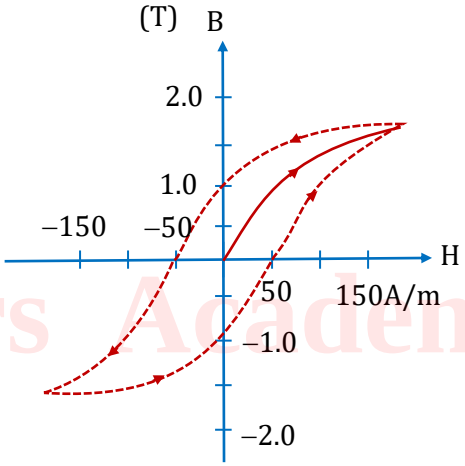
PMG043

10. A rectangular coil (Dimension 5 cm × 2.5 cm) with 100 turns, carrying a current of 3 A in the clock-wise direction is kept centered at the origin and in the X-Z plane. A magnetic field of 1 T is applied along X-axis. If the coil is tilted through 45° about Z-axis, then the torque on the coil is : **[JEE (Main) 2019]**

(1) 0.55 Nm (2) 0.27 Nm (3) 0.38 Nm (4) 0.42 Nm

PMG044

11. The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are: **[JEE (Main) 2020]**



(1) 150 A/m, 1.0 T and 1.5 T (2) 1.0 T, 50 A/m and 1.5 T

(3) 1.5 T, 50 A/m and 1.0 T (4) 1.5 T, 50 A/m and 1.0 T

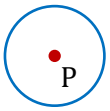
PMG045

12. A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be : **[JEE (Main) 2020]**

(1) 4 A/m (2) 0.75 A/m (3) 2.25 A/m (4) 1 A/m

PMG046

13. A perfectly dimagnetic sphere has a small spherical cavity at its centre, which is filled with a paramagnetic substance. The whole system is placed in a uniform magnetic field $\vec{B}$. Then the field inside the paramagnetic substance is: **[JEE (Main) 2020]**



(1) Zero (2) $\vec{B}$

(3) much large than $|\vec{B}|$ but opposite to $\vec{B}$ (4) much large than $|\vec{B}|$ and parallel to $\vec{B}$

PMG047

Rankers Academy JEE
For More Join: @IITJEE_Advanced

14. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{\imath}$ and magnetic field $\vec{B} = B_0\hat{\imath}$ act on the particle, its speed will double after a time: **[JEE (Main) 2020]**

(1) $\frac{2mv_0}{qE_0}$ (2) $\frac{3mv_0}{qE_0}$ (3) $\frac{\sqrt{3}mv_0}{qE_0}$ (4) $\frac{\sqrt{2}mv_0}{qE_0}$

PMG048

15. The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is :

[JEE (Main) 2022]

(1) $B/2$ (2) $2B$ (3) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (4) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

PMG050

16. The current required to be passed through a solenoid of 15 cm length and 60 turns in order to demagnetise a bar magnet of magnetic intensity $2.4 \times 10^3 Am^{-1}$ is _____ A.

[JEE (Main) 2023]

PMG189

Rankers Academy JEE

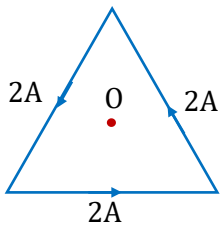
17. A square loop of area $25cm^2$ has a resistance of 10Ω. The loop is placed in uniform magnetic field of magnitude 40.0 T. The plane of loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be

[JEE (Main) 2023]

(1) $2.5 \times 10^{-3}\,J$ (2) $1.0 \times 10^{-3}\,J$ (3) $1.0 \times 10^{-4}\,J$ (4) $5 \times 10^{-3}\,J$

PMG190

18. As shown in the figure, a current of $2A$ flowing in an equilateral triangle of side $4\sqrt{3}cm$. The magnetic field at the centroid O of the triangle is :



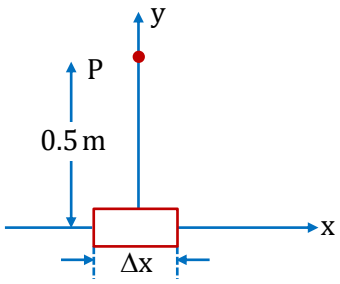
(Neglect the effect of earth's magnetic field.) **[JEE (Main) 2023]**

(1) $4\sqrt{3} \times 10^{-4}T$ (2) $4\sqrt{3} \times 10^{-5}T$ (3) $\sqrt{3} \times 10^{-4}T$ (4) $3\sqrt{3} \times 10^{-5}T$

PMG191

For More Join: @IITJEE_Advanced

19. An element $\Delta l = \Delta x\hat{i}$ is placed at the origin and carries a large current $I = 10A$. The magnetic field on the y-axis at a distance of 0.5 m from the elements Δx of 1 cm length is :



[JEE (Main) 2024]

(1) $4\times10^{-8}T$ (2) $8\times10^{-8}T$ (3) $12\times10^{-8}T$ (4) $10\times10^{-8}T$

PMG212

20. Consider a long thin conducting wire carrying a uniform current I. A particle having mass "M" and charge "q" is released at a distance "a" from the wire with a speed v_o along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability]

[JEE (Main) 2025]

(1) $a\left[1-\frac{mv_o}{2q\mu_o I}\right]$ (2) $\frac{a}{2}$ (3) $a\left[1-\frac{mv_o}{q\mu_o I}\right]$ (4) $ae^{\frac{-4\pi mv_o}{q\mu_o I}}$

PMG263

Rankers Academy JEE

For More Join: @IITJEE_Advanced

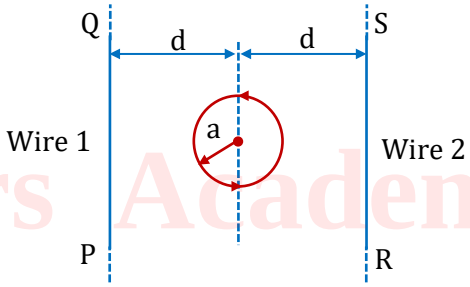
EXERCISE - JEE (Advanced) PYQ

1. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1. In contrast, if the currents I in the two wires have directions opposite to each other, the radius of curvature of the path is R_2. If $\frac{X_0}{X_1}=3$, and value of $\frac{R_1}{R_2}$ is:- **[JEE (Advanced) 2014]**

PMG069

PARAGRAPH FOR QUESTION NO. 2 & 3

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d. The loop and the wires are carrying the same current I. The current in the loop is in the counter clockwise direction if seen from above. **[JEE (Advanced) 2014]**



Rankers Academy JEE

2. When $d \approx a$ but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case

(A) current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx a$

(B) current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx a$

(C) current in wire 1 and wire 2 is the direction PQ and SR, respectively and $h \approx 1.2\ a$

(D) current in wire 1 and wire 2 is the direction PQ and RS, respectively and $h \approx 1.2\ a$

PMG070

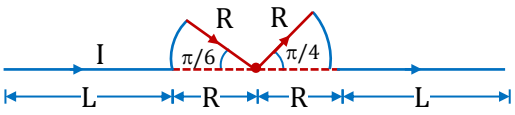
3. Consider $d >> a$, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

(A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

PMG071

For More Join: @IITJEE_Advanced

4. A conductor (shown in the figure) carrying constant current I is kept in the x-y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is (are) **[JEE (Advanced) 2015]**



(A) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$

(B) If $\vec{B}$ is along $\hat{x}$, $F = 0$

(C) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$

(D) If $\vec{B}$ is along $\hat{z}$, $F = 0$

PMG072

Answer Q.5, Q.6 and Q.7 by appropriately matching the information given in the three columns of the following table.

A charged particle (electron or proton) is introduced at the origin (x = 0, y = 0, z = 0) with a given initial velocity $\vec{v}$. A uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ exist everywhere. The velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in column 1, 2 and 3, respectively. The quantities E_0, B_0 are positive in magnitude. **[JEE (Advanced) 2017]**

Column-1		Column-2		Column-3	
(I)	Electron with $\vec{v} = 2\frac{E_0}{B_0}\hat{x}$	(i)	$\vec{E} = E_0\hat{z}$	(P)	$\vec{B} = -B_0\hat{x}$
(II)	Electron with $\vec{v} = \frac{E_0}{B_0}\hat{y}$	(ii)	$\vec{E} = -E_0\hat{y}$	(Q)	$\vec{B} = B_0\hat{x}$
(III)	Proton with $\vec{v} = 0$	(iii)	$\vec{E} = -E_0\hat{x}$	(R)	$\vec{B} = B_0\hat{y}$
(IV)	Proton with $\vec{v} = 2\frac{E_0}{B_0}\hat{x}$	(iv)	$\vec{E} = E_0\hat{x}$	(S)	$\vec{B} = B_0\hat{z}$

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5. In which case will the particle move in a straight line with constant velocity ?

(A) (II) (iii) (S) (B) (IV) (i) (S) (C) (III) (ii) (R) (D) (III) (iii) (P)

PMG073

6. In which case will the particle describe a helical path with axis along the positive z-direction ?

(A) (II) (ii) (R) (B) (IV) (ii) (R) (C) (IV) (i) (S) (D) (III) (iii) (P)

PMG074

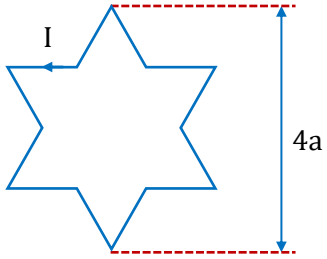
7. In which case would the particle move in a straight line along the negative direction of y-axis (i.e., move along $-\hat{y}$) ?

(A) (IV) (ii) (S) (B) (III) (ii) (P) (C) (II) (iii) (Q) (D) (III) (ii) (R)

PMG075

For More Join: @IITJEE_Advanced

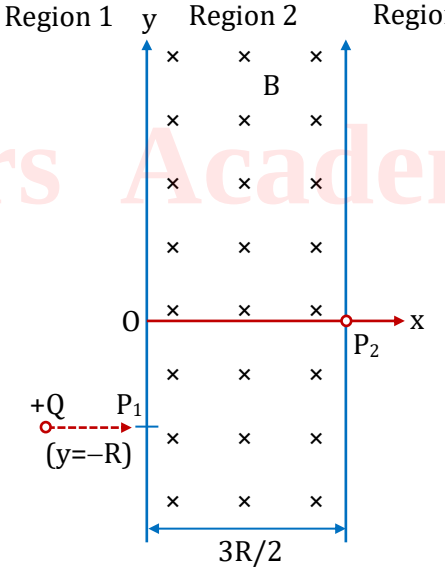
8. A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is $4a$. The magnitude of the magnetic field at the center of the loop is : **[JEE (Advanced) 2017]**



(A) $\frac{\mu_0 I}{4\pi a} 3\left[\sqrt{3}-1\right]$ (B) $\frac{\mu_0 I}{4\pi a} 6\left[\sqrt{3}-1\right]$ (C) $\frac{\mu_0 I}{4\pi a} 6\left[\sqrt{3}+1\right]$ (D) $\frac{\mu_0 I}{4\pi a} 3\left[2-\sqrt{3}\right]$

PMG076

9. A uniform magnetic field B exists in the region between $x = 0$ and $x = \frac{3R}{2}$ (region 2 in the figure) pointing normally into the plane of the paper. A particle with charge $+Q$ and momentum p directed along x-axis enters region 2 from region 1 at point $P_1(y = -R)$. Which of the following options(s) is/are correct ? **[JEE (Advanced) 2017]**



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(A) For $B = \frac{8}{13}\frac{p}{QR}$, the particle will enter region 3 through the point P_2 on x-axis.

(B) For $B > \frac{2}{3}\frac{p}{QR}$, the particle will re-enter region 1.

(C) For a fixed B, particle of same charge Q and same velocity v, the distance between the point P_1 and the point of re-entry into region 1 is inversely proportional to the mass of the particle.

(D) When the particle re-enters region 1 through the longest possible path in region 2, the magnitude of the change in its linear momentum between point P_1 and the farthest point from y-axis is $\frac{p}{\sqrt{2}}$.

PMG077

For More Join: @IITJEE_Advanced

10. Two infinitely long straight wires lie in the xy-plane along the lines $x = \pm R$. The wire located at $x = +R$ carries a constant current I_1 and the wire located at $x = -R$ carries a constant current I_2. A circular loop of radius R is suspended with its centre at $(0, 0, \sqrt{3}R)$ and in a plane parallel to the xy-plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive if it is in the $+\hat{j}$ direction. Which of the following statements regarding the magnetic field $\vec{B}$ is (are) true ? **[JEE (Advanced) 2018]**

(A) If $I_1 = I_2$, then $\vec{B}$ cannot be equal to zero at the origin $(0, 0, 0)$

(B) If $I_1 > 0$ and $I_2 < 0$, then $\vec{B}$ can be equal to zero at the origin $(0, 0, 0)$

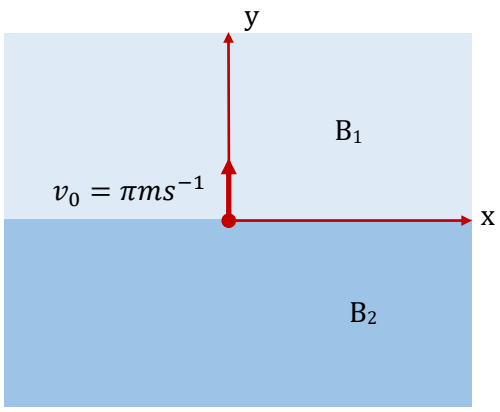
(C) If $I_1 < 0$ and $I_2 > 0$, then $\vec{B}$ can be equal to zero at the origin $(0, 0, 0)$

(D) If $I_1 = I_2$, then the z-component of the magnetic field at the centre of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

PMG078

11. In the x-y-plane, the region $y > 0$ has a uniform magnetic field $B_1\hat{k}$ and the region $y < 0$ has a another uniform magnetic field $B_1\hat{k}$. A positively charged particle is projected from the origin along the positive y-axis with speed $v_0 = \pi ms^{-1}$ at $t = 0$, as shown in the figure. Neglect gravity in this problem. Let $t = T$ be the time when the particle crosses the x-axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1}, along the x-axis in the time interval T is _____.

[JEE (Advanced) 2018]

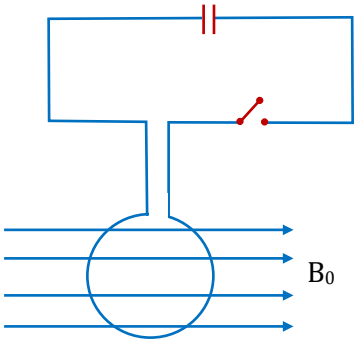


PMG079

For More Join: @IITJEE_Advanced

12. A circular coil of radius R and N turns has negligible resistance. As shown in the schematic figure, its two ends are connected to two wires and it is hanging by those wires with its plane being vertical. The wires are connected to a capacitor with charge Q through a switch. The coil is in a horizontal uniform magnetic field B_o parallel to the plane of the coil. When the switch is closed, the capacitor gets discharged through the coil in a very short time. By the time the capacitor is discharged fully, magnitude of the angular momentum gained by the coil will be (assume that the discharge time is so short that the coil has hardly rotated during this time)-

[JEE (Advanced) 2020]



(A) $\frac{\pi}{2}NQB_oR^2$ (B) πNQB_oR^2 (C) $2\pi NQB_oR^2$ (D) $4\pi NQB_oR^2$

PMG080

13. An α-particle (mass 4 *amu*) and a singly charged sulfur ion (mass 32 *amu*) are initially at rest. They are accelerated through a potential V and then allowed to pass into a region of uniform magnetic field which is normal to the velocities of the particles. Within this region, the α-particle and the sulfur ion move in circular orbits of radii r_α and r_s, respectively. The ratio (r_s/r_α) is___.

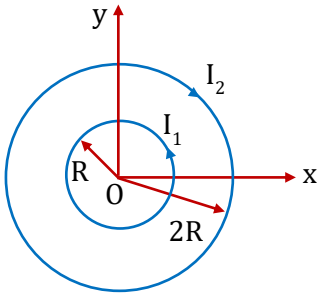
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[JEE (Advanced) 2021]

PMG081

14. Two concentric circular loops, one of radius R and the other of radius $2R$, lie in the xy-plane with the origin as their common center, as shown in the figure. The smaller loop carries current I_1 in the anti-clockwise direction and the larger loop carries current I_2 in the clockwise direction, with $I_2 > 2I_1$. $\vec{B}(x, y)$ denotes the magnetic field at a point (x, y) in the xy-plane. Which of the following statement(s) is(are) correct?

[JEE (Advanced) 2021]



(A) $\vec{B}(x,y)$ is perpendicular to the xy-plane at any point in the plane.

(B) $|\vec{B}(x,y)|$ depends on x and y only through the radial distance $r = \sqrt{x^2+y^2}$.

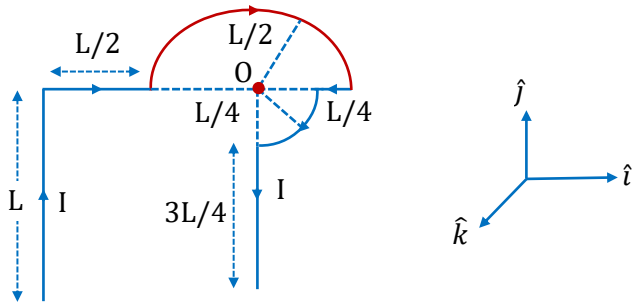
(C) $|\vec{B}(x,y)|$ is non-zero at all points for $r < R$.

(D) $\vec{B}(x,y)$ points normally outward from the xy-plane for all the points between the two loops.

PMG082

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15. Which one of the following options represents the magnetic field $\vec{B}$ at O due to the current flowing in the given wire segments lying on the xy plane? **[JEE (Advanced) 2022]**



(A) $\vec{B} = \frac{-\mu_0 I}{L}\left(\frac{3}{2} + \frac{1}{4\sqrt{2}\pi}\right)\hat{k}$

(B) $\vec{B} = -\frac{\mu_0 I}{L}\left(\frac{3}{2} + \frac{1}{2\sqrt{2}\pi}\right)\hat{k}$

(C) $\vec{B} = \frac{-\mu_0 I}{L}\left(1 + \frac{1}{4\sqrt{2}\pi}\right)\hat{k}$

(D) $\vec{B} = \frac{-\mu_0 I}{L}\left(1 + \frac{1}{4\pi}\right)\hat{k}$

PMG083

16. An infinitely long wire, located on the z-axis, carries a current I along the +z-direction and produces the magnetic field $\vec{B}$. The magnitude of the line integral $\int \vec{B}.\vec{dl}$ along a straight line from the point $(-\sqrt{3}a, a, 0)$ to $(a, a, 0)$ is given by

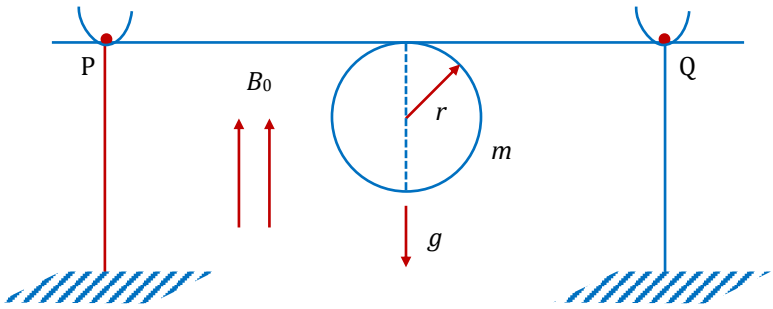
[μ_0 is the magnetic permeability of free space.] **[JEE (Advanced) 2024]**

(A) $7\mu_0 I/24$ (B) $7\mu_0 I/12$ (C) $\mu_0 I/8$ (D) $\mu_0 I/6$

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PMG213

17. A thin stiff insulated metal wire is bent into a circular loop with its two ends extending tangentially from the same point of the loop. The wire loop has mass m and radius r and it is in a uniform vertical magnetic field B_0, as shown in the figure. Initially, it hangs vertically downwards, because of acceleration due to gravity g, on two conducting supports at P and Q. When a current I is passed through the loop, the loop turns about the line PQ by an angle θ given by



[JEE (Advanced) 2024]

(A) $\tan\theta = \pi r I B_0/(mg)$

(B) $\tan\theta = 2\pi r I B_0/(mg)$

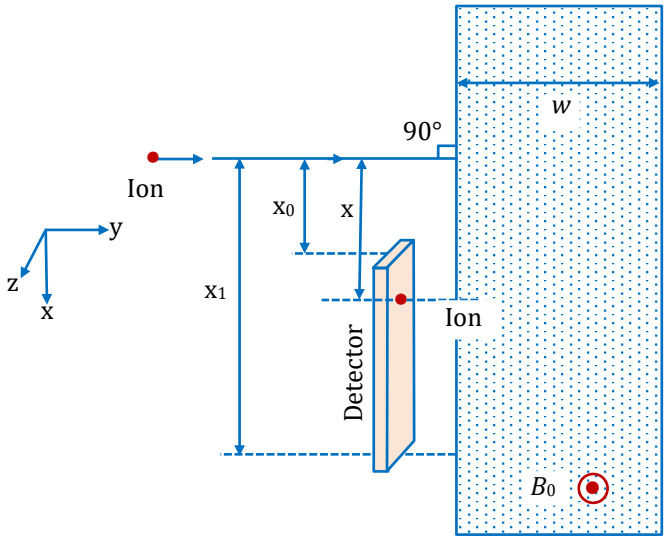
(C) $\tan\theta = \pi r I B_0/(2mg)$

(D) $\tan\theta = mg/(\pi r I B_0)$

PMG214

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18. A positive, singly ionized atom of mass number A_M is accelerated from rest by the voltage 192 V. Thereafter, it enters a rectangular region of width w with magnetic field $\vec{B}_0 = 0.1\hat{k}$ Tesla, as shown in the figure. The ion finally hits a detector at the distance x below its starting trajectory.
[Given: Mass of neutron/proton = $(5/3) \times 10^{-27}$ kg, charge of the electron = 1.6×10^{-19} C.]



Which of the following option(s) is(are) correct? **[JEE (Advanced) 2024]**

(A) The value of x for H^+ ion is 4 cm.

(B) The value of x for an ion with $A_M = 144$ is 48 cm.

(C) For detecting ions with $1 \le A_M \le 196$, the minimum height $(x_1 - x_0)$ of the detector is 55 cm.

(D) The minimum width w of the region of the magnetic field for detecting ions with $A_M = 196$ is 56 cm.

PMG215

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	B	C	A	A	A	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	D	A	D	A	C	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	A	C	A	B	B	A	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	D	A	B	D	D	B	A	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	B	D	A	B	C	D	C	B	B
Que.	51	52								
Ans.	C	A								

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ACD	AD	ACD	AC	CD	ABC	AB	ABCD	ABD	AC
Que.	11	12	13	14	15	16	17	18	19	
Ans.	ABCD	ABD	A	B	B	C	B	B	B	
Que.	20		21							
Ans.	(A-Q,S), (B-P,S), (C-Q,S), (D-P,S)		(A-P,R,T), (B-P,Q), (C-S), (D-P)							
Que.	22									
	(A-S), (B-S), (C-Q), (D-R), (E-P)									

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EXERCISE (S)

1. 1 **2.** $B=\frac{1}{2}\mu_0 K$ **3.** 18

4. 5 **5.** $7\mu T$ **6.** $\frac{\mu_0 I\sqrt{2}}{4\pi^2 R}$ **7.** $\frac{\pi+2\theta}{\pi-2\theta}$

8. 5 **9.** 5

10. (a) A horizontal magnetic field of magnitude 0.26 *T* normal to the conductor in such a direction that Fleming's left-hand rule gives a magnetic force upward.

(b) 1.176 *N*.

11. 1 **12.** Use $\tau = I(A \times B)$ and $F = I(l \times B)$ **13.** $T_0=2\pi\sqrt{\frac{m}{6IB}}=0.57s$

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	3	2	3	4	1	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	1	3	3	6	2	4	1	4

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	C	B	ABC	A	C	D	B	AB	ABD
Que.	11			12	13	14	15	16	17	18
Ans.	(2) [1.99, 2.01]			B	4	AB	C	A	A	AB

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